

Chapter: 3

list: ordered n tuple

$$(a, b, c, c, d) \neq (a, c, b, c, d)$$

list: collection of objects where the sequence matters
length: no of elements of a list

$$(b, 0, 5): 3$$

When can we say two lists are equal?

→ two lists are equal if they have the same elements at exactly the same position.

$$(a, b, c, d) = (x, p, q, s)$$

$$\therefore x = a, p = b, q = c, s = d$$

$()$: empty list (list with no elements)

$$(a, b, c, d, e, f) \rightarrow (), \text{ with}$$

abcdef : string

bag: $\square \square$

$$(2 \square, \square, 2 \square, \square, 2 \square, \square, \dots)$$

$$\square \square, \square \square, \square \square, \dots$$

Multiplication principle:

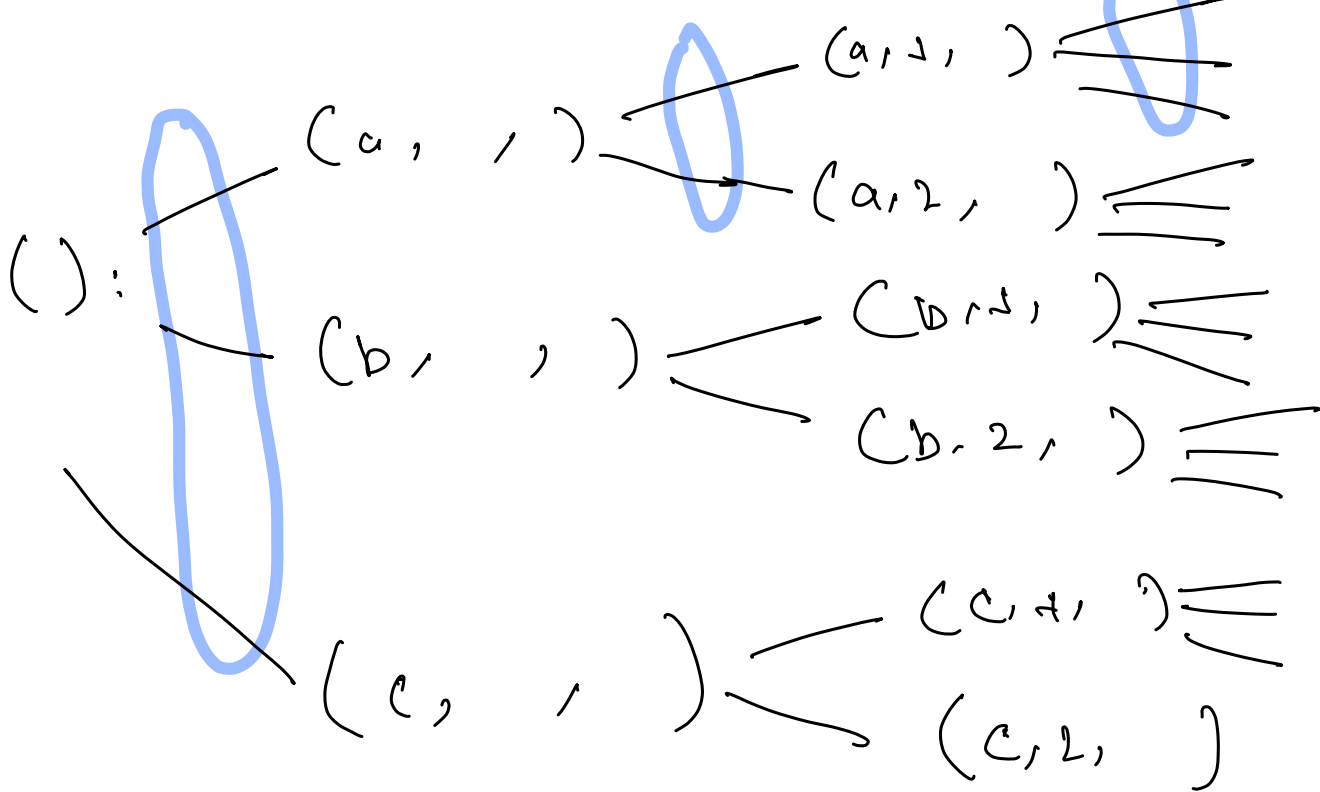
we are trying to make a list of length: 3



choice of A

choice of B

choice of C

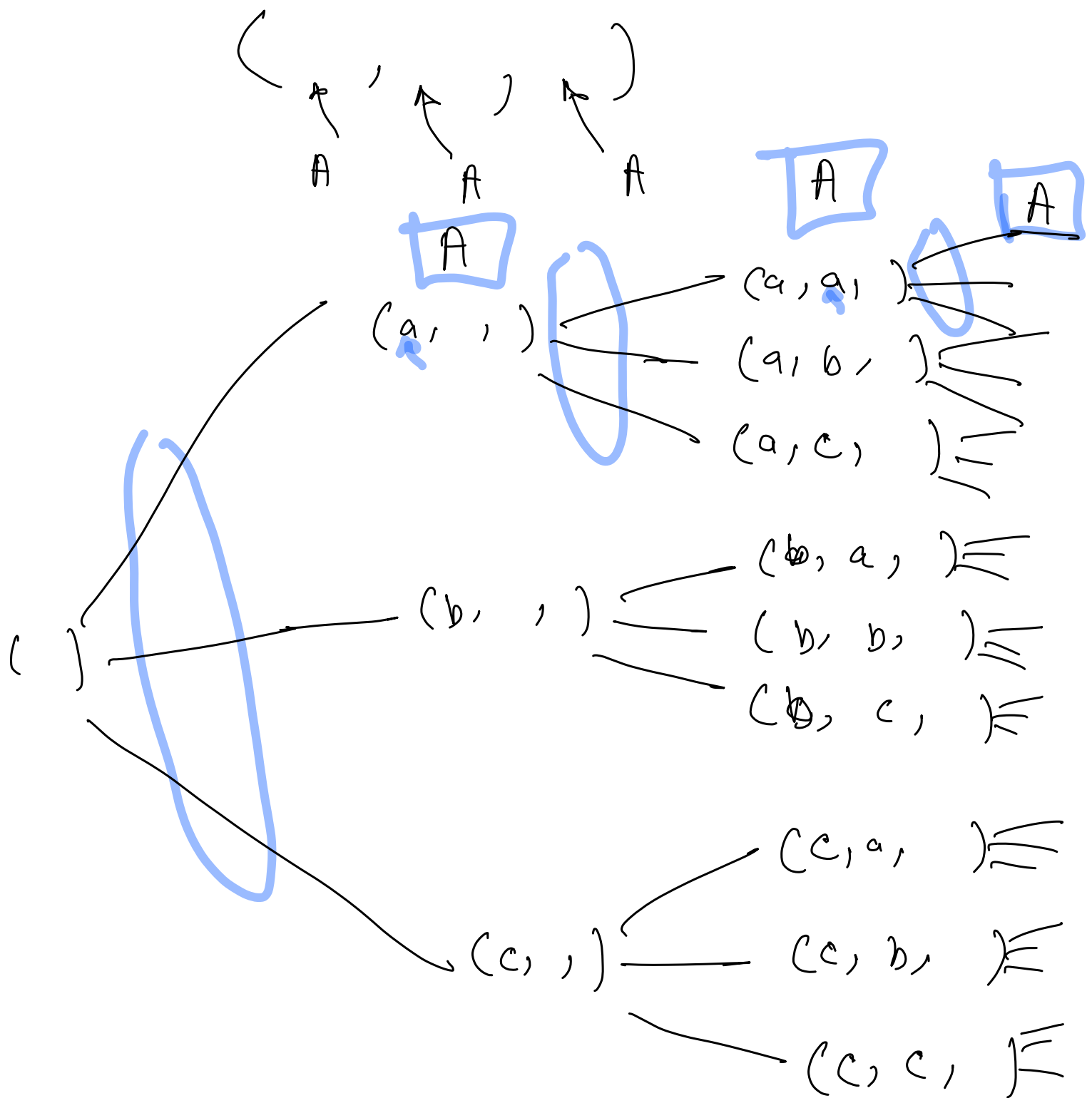


$$\text{Total: } 3 \cdot 2 \cdot 3 = 18$$

we created a list:



$$A = \{a, b, c\}$$

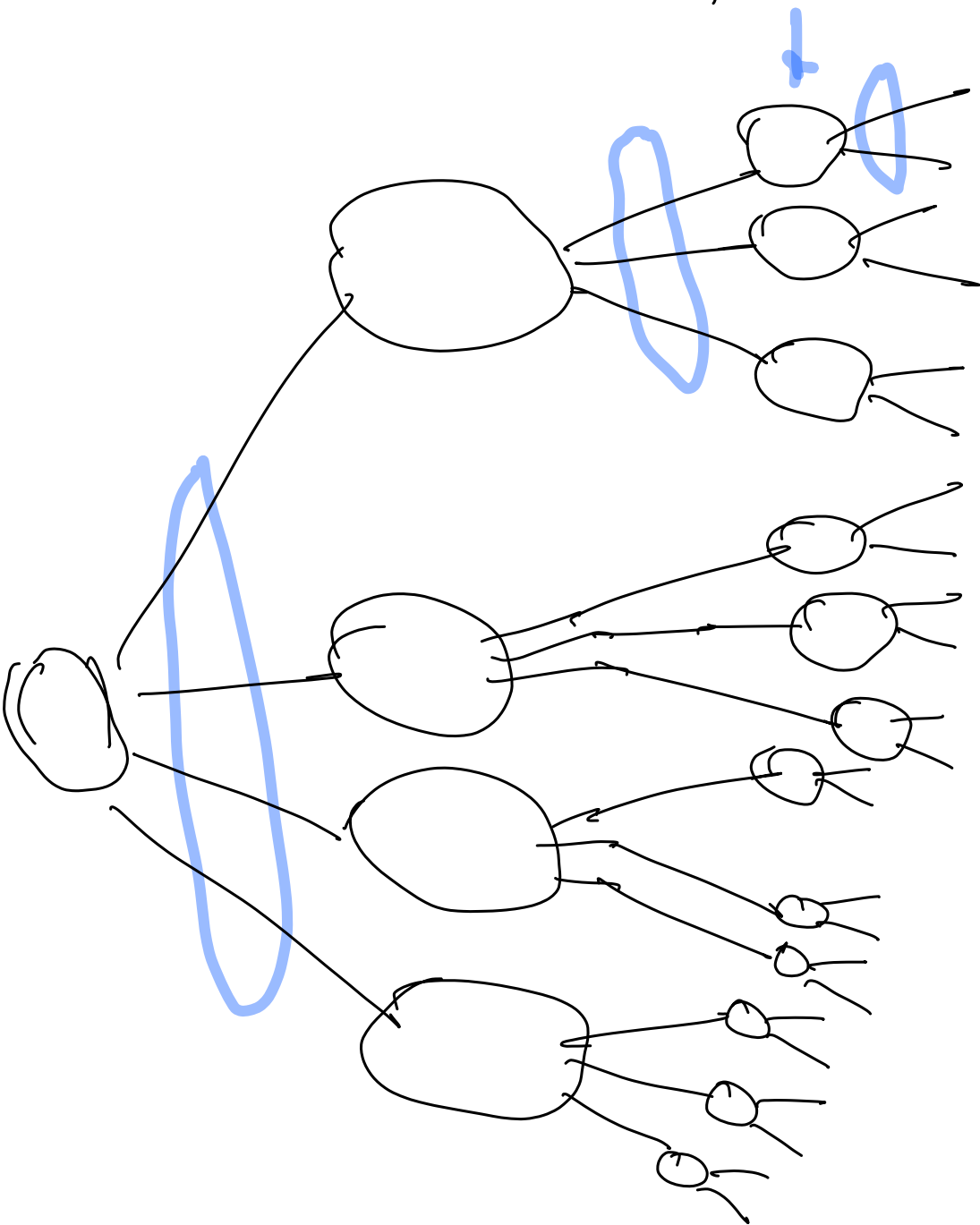


$$\text{Total} = 3^3$$

[we are allowing repetition]

$(\frac{\square}{\square}, \frac{A}{\square}, -)$
 $\uparrow \quad \quad \uparrow \quad \quad \uparrow$
 $A \quad A \setminus \{a\} \quad A \setminus \{a, A\}$

$A = \{a, b, c, d\}$



Total = $4 \cdot 3 \cdot 2$

Multiplication principle:

list a of n elements:

for each slot, I have $a_1, a_2, \dots, a_n$

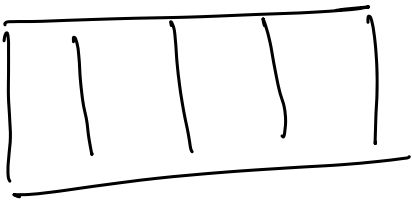
no of choices

Then the total no of list constructed is:

$$a_1 \cdot a_2 \cdot a_3 \cdot \dots \cdot a_n$$

example:

$$A = \{a, b, c, d\}$$



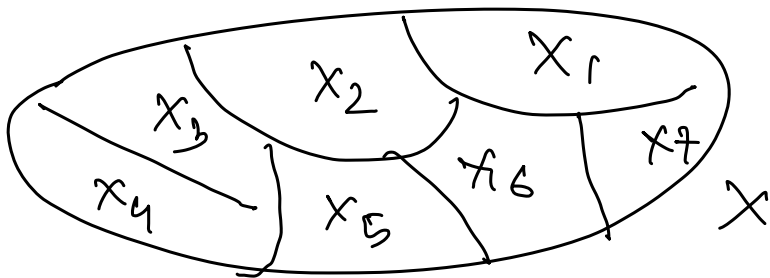
repetition: 4^4

non-repetition: $4 \cdot 3 \cdot 2 \cdot 1$

Additive principle:

If a finite X can be decomposed into disjoint set X_i

$$\text{then, } |X| = |X_1| + |X_2| + |X_3| + \dots + |X_n|$$



$$X_i \cap X_j = \emptyset$$

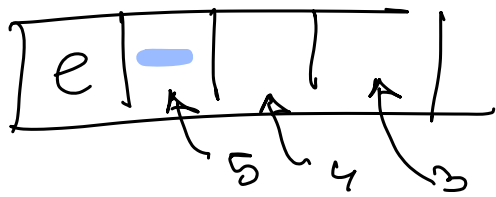
example: $A = \{a, b, c, d, e, f\}$ is a choice set,

you have to create a list of size: 4

such that, e is always selected, (repetition of size 4 not allowed)

X : all possible lists containing e :

X_1 : all possible list of size 4 where e is 1st element



$A = \{a, b, c, d, e, f\} \setminus \{e\}$

X_2 : all possible list

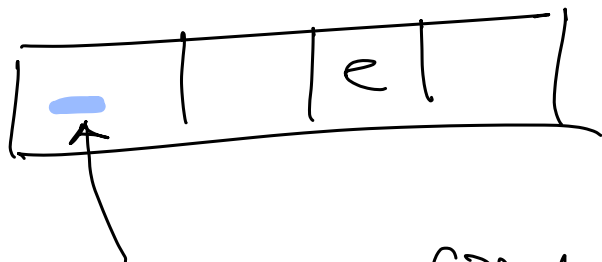
where e is 2nd element



$= 5 \cdot 4 \cdot 3$

$A = \{a, b, c, d, e, f\} \setminus \{e\}$

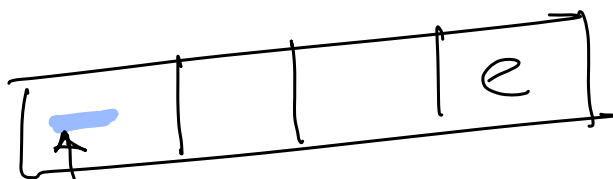
X_3 :



$= 5 \cdot 4 \cdot 3$

$A = \{a, b, c, d, e, f\} \setminus \{e\}$

X_4 :



$= 5 \cdot 4 \cdot 3$

$A = \{a, b, c, d, e, f\} \setminus \{e\}$

$$|X| = |X_1| + |X_2| + |X_3| + |X_4|$$

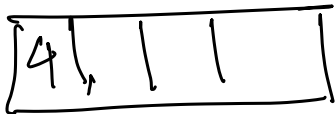
$$= 5 \cdot 4 \cdot 3 + 5 \cdot 4 \cdot 3 + 5 \cdot 4 \cdot 3 + 5 \cdot 4 \cdot 3$$

$$= 240$$

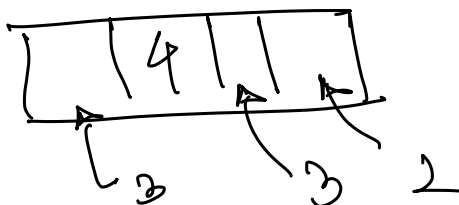
ex: $A = \{6, 3, 0, 4, 5\}$

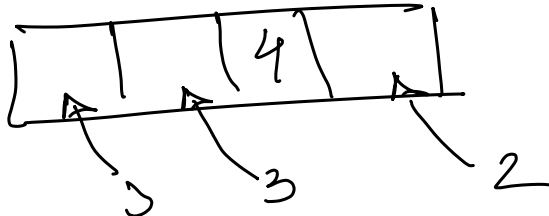
construct a 4 digit number where 4 is always selected,

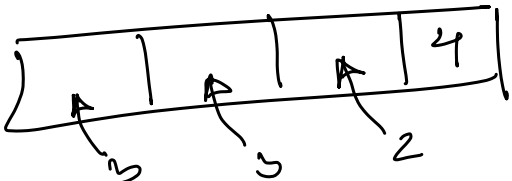
X : all possible 4 digit number containing 4

X_1 :  $\Rightarrow 4 \cdot 3 \cdot 2 = 24$

$A = \{6, 3, 0, 4, 5\}$

X_2 :  $\Rightarrow 3 \cdot 3 \cdot 2 = 18$

X_3 :  $\Rightarrow 3 \cdot 3 \cdot 2 = 18$

X_4 :  $\Rightarrow 3 \cdot 3 \cdot 2 = 18$

$$|X| = 18 + 18 + 18 + 24 = 78$$

subtraction principle;

if X is a subset of a finite set U then

$$|\bar{X}| = |U| - |X|$$

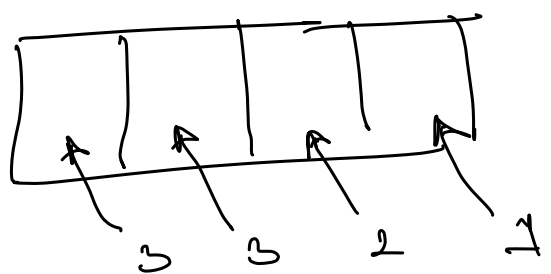
in other words, if $X \subseteq U$, then $|\bar{X}| = |U - X|$
 $= |U| - |X|$

example:

ex: $A = \{6, 3, 0, 4, 5\}$

construct a 4 digit number where 4 is always selected,

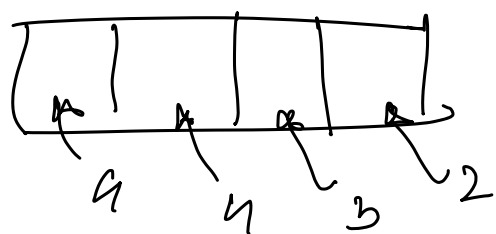
X : string of 4 digit numbers without 4



$$A = \{6, 3, 0, 4, 5\}$$

$$|X| = 3 \cdot 3 \cdot 2 \cdot 1 = 18$$

Total: 4 digit numbers



$$|U| = 4 \cdot 4 \cdot 3 \cdot 2 = 96$$

$\bar{X}$ = string of digit numbers containing 4

$$\therefore |\bar{X}| = |U| - |X| = 96 - 18 = 78$$

Division principle:

floor: $\lfloor x \rfloor$: rounded down

ceiling: $\lceil x \rceil$: rounded up

$$\lfloor 13.2 \rfloor = 13 \quad \lceil 13.2 \rceil = 14$$

Principle:

n objects are placed into k boxes

- at least one box containing $\lceil \frac{n}{k} \rceil$ or more objects.
- at least one box containing $\lfloor \frac{n}{k} \rfloor$ or fewer objects
- if $n > k$, then $\frac{n}{k} > 1$ so, $\lceil \frac{n}{k} \rceil > 1$

some boxes contain more than 1 objects

- if $n < k$ then $\frac{n}{k} < 1$, so, $\lfloor \frac{n}{k} \rfloor < 1$

at least one of the boxes are empty.

pigeon hole principle,

example: a group of 13 people:

There is a month on which at least two people have birthdays

Direct proof:

Theorem: a mathematical statement which is true, and can be proven true.

proof: a formal / written verification (process) that shows a theorem is true,

Theorem: if $\sum_{k=1}^{\infty} a_k$ converges then $\lim_{k \rightarrow \infty} a_k = 0$

Theorem: f is continuous on the interval $[a, b]$ it is Riemann integrable,

If P then Q , $P \Rightarrow Q$

Theorem: Bidirectional arrow: $P \Leftrightarrow Q$

in $\mathbb{R}$ a sequence is a Cauchy sequence if and only if it is a convergent sequence